

EXP.NO: 9	Contiguous Memory Allocation – First Fit and Best Fit
DATE: 20/03/2026	

AIM:

To implement Contiguous Memory Allocation using First Fit and Best Fit algorithms.

ALGORITHM:

- First Fit
 1. Start from first memory block.
 2. Allocate first block that fits the process.
 3. Repeat for all processes.
- Best Fit
 1. Search all memory blocks.
 2. Allocate smallest block that fits the process.
 3. Repeat for all processes.

CODE:

```
#include<stdio.h>
int main()
{
    int b[10], p[10], i, j, nb, np;
    printf("Enter number of blocks: ");
    scanf("%d",&nb);
    printf("Enter block sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&b[i]);
    printf("Enter number of processes: ");
    scanf("%d",&np);
    printf("Enter process sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&p[i]);
    printf("\nFirst Fit Allocation\n");
    for(i=0;i<np;i++)
    {
        for(j=0;j<nb;j++)
        {
            if(b[j] >= p[i])
            {
                printf("Process %d allocated to block %d\n", i+1, j+1);
                b[j] = b[j] - p[i];
                break;
            }
        }
    }
}
```

OUTPUT:

```
[hp@localhost ~]$ gcc memory_allocation.c -o memory
[hp@localhost ~]$ ./memory
"
CONTIGUOUS MEMORY ALLOCATION
=====
Enter number of blocks: 5
Enter block sizes: 100 500 200 300 600
Enter number of processes: 4
Enter process sizes: 212 417 112 426

FIRST FIT ALLOCATION
-----
Process No.    Process Size    Block No.    Block Size    Internal Fragment
1              212             2             500           288
2              417             5             600           183
3              112             3             200           88
4              426            Not Allocated
Total Internal Fragmentation (First Fit) = 559

BEST FIT ALLOCATION
-----
Process No.    Process Size    Block No.    Block Size    Internal Fragment
1              212             3             300           88
2              417             2             500           83
3              112             1             100           -
4              426             5             600          174
Total Internal Fragmentation (Best Fit) = 345

[hp@localhost ~]$ █
```

RESULT:

Thus, Contiguous Memory Allocation using First Fit and Best Fit algorithms was implemented and executed successfully.